

Topics: 16.4 Green's Theorem; 16.5 Curl and Divergence; 16.6 Parametric Surfaces and Their Areas

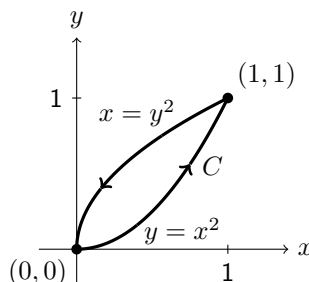
Please put away all electronic devices, including cell phones and calculators.

This content is protected and may not be shared, uploaded, or distributed.

1. Use Green's Theorem to evaluate the line integral.

- (a) $\oint_C (y + e^{\sqrt[3]{x^5}}) dx + (2x + \cos(y^2)) dy$, where C is the positively oriented boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

We first draw the curve C .



Let $P(x, y) = y + e^{\sqrt[3]{x^5}}$ and $Q(x, y) = 2x + \cos(y^2)$. By Green's Theorem,

$$\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA.$$

Taking partial derivatives, we find that $Q_x = 2$ and $P_y = 1$. Note that the region bounded by C is

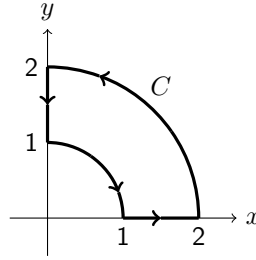
$$D = \{(x, y) \mid 0 \leq x \leq 1, x^2 \leq y \leq \sqrt{x}\}.$$

Thus,

$$\begin{aligned} \oint_C (y + e^{\sqrt[3]{x^5}}) dx + (2x + \cos(y^2)) dy &= \iint_D (2 - 1) dA \\ &= \int_0^1 \int_{x^2}^{\sqrt{x}} 1 dy dx \\ &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left[\frac{2}{3} x^{3/2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{3}. \end{aligned}$$

- (b) $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (3 + e^{x^2})\mathbf{i} + (\arctan y + 3x^2)\mathbf{j}$ and C is the positively oriented boundary of the quarter annulus $1 \leq x^2 + y^2 \leq 4$ in the first quadrant.

We first draw the boundary curve C .



Let $P(x, y) = 3 + e^{x^2}$ and $Q(x, y) = \arctan y + 3x^2$. By Green's Theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA,$$

where D is the region bounded by C . We find that $Q_x = 6x$ and $P_y = 0$. Thus,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_D 6x dA \\ &= \int_0^{\pi/2} \int_1^2 6r^2 \cos \theta dr d\theta \\ &= 6 \int_0^{\pi/2} \cos \theta \left[\frac{r^3}{3} \right]_1^2 d\theta \\ &= 14 \int_0^{\pi/2} \cos \theta d\theta \\ &= 14[\sin \theta]_0^{\pi/2} = 14. \end{aligned}$$

2. Consider the force field $\mathbf{F}(x, y) = xy^2 \mathbf{i} + 5x^2y \mathbf{j}$.

(a) Determine whether $\mathbf{F}$ is conservative.

We have that $\mathbf{F} = \langle P, Q \rangle$, where $P(x, y) = xy^2$ and $Q(x, y) = 5x^2y$. Taking partial derivatives,

$$P_y = 2xy, \quad Q_x = 10xy.$$

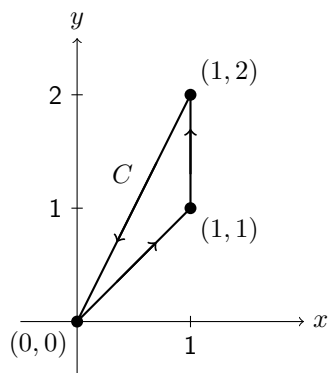
Since $P_y \neq Q_x$, the vector field is not conservative.

(b) Find the work done by $\mathbf{F}$ in moving a particle around the positively oriented triangular path C with vertices $(0, 0)$, $(1, 1)$, and $(1, 2)$.

The work done by $\mathbf{F}$ in moving a particle along the closed path C is given by

$$\text{Work} = \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

We first draw the curve C .



Let D be the triangular region bounded by C . By Green's Theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA.$$

Taking partial derivatives, we find that $Q_x = 10xy$ and $P_y = 2xy$. Thus,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \int_x^{2x} (10xy - 2xy) dy dx \\ &= \int_0^1 \int_x^{2x} 8xy dy dx \\ &= \int_0^1 [4xy^2]_x^{2x} dx \\ &= \int_0^1 12x^3 dx \\ &= [3x^4]_0^1 = 3. \end{aligned}$$

3. Compute the curl and divergence of each vector field.

(a) $\mathbf{F}(x, y, z) = x^3yz^2\mathbf{j} + y^4z^3\mathbf{k}$

We have that

$$\begin{aligned} \text{curl } \mathbf{F} &= \nabla \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x^3yz^2 & y^4z^3 \end{vmatrix} \\ &= (4y^3z^3 - 2x^3yz)\mathbf{i} - (0 - 0)\mathbf{j} + (3x^2yz^2 - 0)\mathbf{k} \\ &= (4y^3z^3 - 2x^3yz)\mathbf{i} + 3x^2yz^2\mathbf{k}. \end{aligned}$$

Also,

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = 0 + x^3z^2 + 3y^4z^2 = x^3z^2 + 3y^4z^2.$$

(b) $\mathbf{F}(x, y, z) = \ln(2y + 3z)\mathbf{i} + \ln(x + 3z)\mathbf{j} + \ln(x + 2y)\mathbf{k}$

We have that

$$\begin{aligned}\operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \ln(2y+3z) & \ln(x+3z) & \ln(x+2y) \end{vmatrix} \\ &= \left(\frac{2}{x+2y} - \frac{3}{x+3z} \right) \mathbf{i} - \left(\frac{1}{x+2y} - \frac{3}{2y+3z} \right) \mathbf{j} + \left(\frac{1}{x+3z} - \frac{2}{2y+3z} \right) \mathbf{k}.\end{aligned}$$

Also,

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = 0 + 0 + 0 = 0.$$

(c) $\mathbf{F}(x, y, z) = (x^2 \ln(y+1))\mathbf{i} + (y^2 z^3)\mathbf{j} + \left(\frac{z}{y}\right)\mathbf{k}$

We have that

$$\begin{aligned}\operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} \\ &= (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k} \\ &= \left(-\frac{z}{y^2} - 3y^2 z^2 \right) \mathbf{i} + (0 - 0)\mathbf{j} + \left(0 - \frac{x^2}{y+1} \right) \mathbf{k} \\ &= \left\langle -\frac{z}{y^2} - 3y^2 z^2, 0, -\frac{x^2}{y+1} \right\rangle.\end{aligned}$$

Also,

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = 2x \ln(y+1) + 2yz^3 + \frac{1}{y}.$$

4. Find a parametric representation for the surface.

(a) The part of the hyperboloid $4x^2 - 4y^2 - z^2 = 4$ that lies in front of the yz -plane.

Solving for x , we get

$$4x^2 - 4y^2 - z^2 = 4 \implies x^2 = y^2 + \frac{z^2}{4} + 1.$$

Since the surface lies in front of the yz -plane, we have $x \geq 0$. Thus

$$x = \sqrt{y^2 + \frac{z^2}{4} + 1}.$$

Letting y and z be the parameters, a parametrization is

$$\mathbf{r}(y, z) = \left\langle \sqrt{y^2 + \frac{z^2}{4} + 1}, y, z \right\rangle, \quad -\infty < y < \infty, \quad -\infty < z < \infty.$$

(b) The part of the cylinder $x^2 + z^2 = 9$ that lies above the xy -plane and between the planes $y = -4$ and $y = 4$.

Since $x^2 + z^2 = 9$, we may write $x = 3 \cos \theta$ and $z = 3 \sin \theta$. The surface lies above the xy -plane, so $z \geq 0$, which gives $0 \leq \theta \leq \pi$. Also, the planes $y = -4$ and $y = 4$ give $-4 \leq y \leq 4$. Therefore, a parametrization is

$$\mathbf{r}(\theta, y) = \langle 3 \cos \theta, y, 3 \sin \theta \rangle, \quad 0 \leq \theta \leq \pi, \quad -4 \leq y \leq 4.$$

5. Find an equation of the tangent plane to the surface at the given point.

(a) The surface given by $x = u^2 + 1$, $y = v^3 + 1$, $z = u + v$ at the point $P(5, 2, 3)$.

The surface is parametrized by

$$\mathbf{r}(u, v) = \langle u^2 + 1, v^3 + 1, u + v \rangle.$$

Taking partial derivatives,

$$\mathbf{r}_u = \langle 2u, 0, 1 \rangle \quad \text{and} \quad \mathbf{r}_v = \langle 0, 3v^2, 1 \rangle.$$

A normal vector to the tangent plane is

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2u & 0 & 1 \\ 0 & 3v^2 & 1 \end{vmatrix} = \langle -3v^2, -2u, 6uv^2 \rangle.$$

Notice that the point $P(5, 2, 3)$ corresponds to $\mathbf{r}(2, 1)$. We have

$$\mathbf{r}_u(2, 1) \times \mathbf{r}_v(2, 1) = \langle -3, -4, 12 \rangle.$$

Thus an equation of the tangent plane is

$$-3(x - 5) - 4(y - 2) + 12(z - 3) = 0.$$

(b) The surface described by $\mathbf{r}(u, v) = u^2 \mathbf{i} + 2u \sin v \mathbf{j} + u \cos v \mathbf{k}$ at the point corresponding to $(u, v) = (1, 0)$.

Taking partial derivatives,

$$\mathbf{r}_u = \langle 2u, 2 \sin v, \cos v \rangle \quad \text{and} \quad \mathbf{r}_v = \langle 0, 2u \cos v, -u \sin v \rangle.$$

At $(u, v) = (1, 0)$,

$$\mathbf{r}_u(1, 0) = \langle 2, 0, 1 \rangle \quad \text{and} \quad \mathbf{r}_v(1, 0) = \langle 0, 2, 0 \rangle.$$

Thus a normal vector to the tangent plane is

$$\mathbf{r}_u(1, 0) \times \mathbf{r}_v(1, 0) = \langle -2, 0, 4 \rangle.$$

Also, $\mathbf{r}(1, 0) = \langle 1, 0, 1 \rangle$, so the point on the surface is $(1, 0, 1)$. Therefore an equation of the tangent plane is

$$-2(x - 1) + 4(z - 1) = 0.$$

6. Find the area of the surface S that is the part of the paraboloid $y = x^2 + z^2$ that lies within

the cylinder $x^2 + z^2 = 16$.

We write the surface as

$$y = g(x, z) = x^2 + z^2.$$

Since the surface lies within the cylinder $x^2 + z^2 = 16$, the projection onto the xz -plane is

$$D = \{(x, z) \mid x^2 + z^2 \leq 16\}.$$

Taking partial derivatives, $g_x = 2x$ and $g_z = 2z$. Thus

$$A(S) = \iint_D \sqrt{1 + g_x^2 + g_z^2} dA = \iint_D \sqrt{1 + 4x^2 + 4z^2} dA.$$

Using polar coordinates in the xz -plane, let

$$x = r \cos \theta, \quad z = r \sin \theta,$$

where $0 \leq r \leq 4$ and $0 \leq \theta \leq 2\pi$. Then

$$\begin{aligned} A(S) &= \int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} r dr d\theta \\ &= 2\pi \int_0^4 r \sqrt{1 + 4r^2} dr. \end{aligned}$$

Let $u = 1 + 4r^2$, so $du = 8r dr$. When $r = 0$, $u = 1$, and when $r = 4$, $u = 65$. Therefore

$$\begin{aligned} A(S) &= 2\pi \int_1^{65} u^{1/2} \frac{du}{8} \\ &= \frac{\pi}{4} \left[\frac{2}{3} u^{3/2} \right]_1^{65} \\ &= \frac{\pi}{6} (65^{3/2} - 1). \end{aligned}$$

Alternatively, we could parametrize the surface by

$$\mathbf{r}(x, z) = \langle x, x^2 + z^2, z \rangle$$

and proceed using the formula for the surface area of a parametric surface.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Green's Theorem. Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane, and let D be the region bounded by C . Assume that P and Q have continuous first-order partial derivatives on an open region containing D . Then

$$\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA.$$

Divergence and Curl. Let

$$\mathbf{F}(x, y, z) = P(x, y, z) \mathbf{i} + Q(x, y, z) \mathbf{j} + R(x, y, z) \mathbf{k}.$$

Assume that all first-order partial derivatives of P , Q , and R exist. The **divergence** of $\mathbf{F}$ is

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = P_x + Q_y + R_z.$$

The **curl** of $\mathbf{F}$ is

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (R_y - Q_z) \mathbf{i} + (P_z - R_x) \mathbf{j} + (Q_x - P_y) \mathbf{k}.$$

Tangent Planes to Parametric Surfaces. Suppose that S is given parametrically by

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}.$$

Let P_0 be a point on S with position vector $\mathbf{r}(u_0, v_0) = \langle x_0, y_0, z_0 \rangle$. Then

$$\mathbf{n} = \mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$$

is a normal vector to the tangent plane of S at P_0 , provided this vector is nonzero.

Surface Area. Suppose that a smooth parametric surface S is given by

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k},$$

where S is covered exactly once as (u, v) ranges over the parameter domain D . The surface area of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA.$$

For the special case where S is given explicitly by $z = f(x, y)$, where $(x, y) \in D$ and f has continuous first partial derivatives, the surface area is

$$A(S) = \iint_D \sqrt{1 + f_x^2 + f_y^2} \, dA,$$

where D is the projection of S onto the xy -plane.

Suggested Textbook Problems

Section 16.4: 1-18, 21-23

Section 16.5: 1-12, 15-20

Section 16.6: 1-26, 33-50